

Dr. A. ANTONY ALOSANAI PROMILTON

Ph.D. · Hydrogeologist · Consultant Geophysicist · Geospatial Researcher

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22
JOURNAL ARTICLES

7
BOOK CHAPTERS

45+
CITATIONS

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JOURNALS REVIEWED

■ PERSONAL INFORMATION

Date of Birth 21 October 1994
Gender Male
Nationality Indian
Category OBC
Marital Status Married
Languages Tamil (Native), English (Fluent)

■ FIELDS OF RESEARCH INTEREST

Hydrogeology and groundwater exploration · Applied geophysics (VES, ASAR, ERT, AMT/ADMT) · Coastal hydrogeology · Hard rock aquifer characterization · Hydrogeochemistry and water quality assessment · GIS, remote sensing and geospatial analysis · MCDA (AHP, TOPSIS) · Machine learning in hydrology · Sediment geochemistry and marine studies · Disaster vulnerability mapping · Sustainable water resource management · Renewable energy site selection

■ ACADEMIC QUALIFICATIONS

Degree	University / Institution	Year	Marks / Grade
Ph.D. (Geology)	Manonmaniam Sundaranar University, Tirunelveli	2025	—
M.Sc. (Geology)	V.O. Chidambaram College, Thoothukudi	2017	79% (Dist.)
B.Sc. (Geology)	V.O. Chidambaram College, Thoothukudi	2015	78% (Dist.)

Ph.D. Thesis: 'Groundwater Investigations Using Azimuthal Square Array Method and Geochemical Modelling in Hard Rock Terrain, Valliyur Region, Tamil Nadu'

Guide: Dr. A. Antony Ravindran, Assistant Professor, V.O. Chidambaram College, Thoothukudi (Affiliated to MSU, Tirunelveli)

■ WORK EXPERIENCE

Position	Organization	From	To	Nature
Consultant Geologist / Geophysicist	Imman Geotech Pvt. Ltd., Thoothukudi	Mar 2021	Present	Freelance / Consultant
Subject Matter Expert – Earth Sciences	Chegg India Pvt. Ltd.	Aug 2018	Mar 2025	Freelance

Hydrogeologist	Ratnam Energy Pvt. Ltd., Gujarat	Jul 2017	Aug 2018	Full-time
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■ RESEARCH EXPERTISE & TECHNICAL SKILLS

i. Hydrogeology & Groundwater Characterization

- ▶ Groundwater flow pattern analysis, aquifer characterization, hydraulic property determination and groundwater storage dynamics
- ▶ Pumping tests, slug tests, recovery analysis for transmissivity and storativity parameter estimation
- ▶ Well drilling supervision, lithological logging, water level monitoring and geochemical sampling

ii. Geophysical Surveying & Subsurface Investigation

- ▶ Vertical Electrical Sounding (VES), Azimuthal Square Array Resistivity (ASAR), ERT, AMT/ADMT surveys
- ▶ Fracture orientation mapping and secondary porosity estimation in hard rock terrain using ASAR
- ▶ Geophysical data interpretation, subsurface lithological modelling, 3D aquifer modelling

iii. Hydrogeochemical Analysis & Water Quality

- ▶ Major ion analysis, trace elements, heavy metal concentration determination
- ▶ WQI, DRASTIC vulnerability index, contamination factor analysis and health risk assessment
- ▶ PCA, HCA and multivariate statistical analysis for hydrogeochemical data interpretation

iv. GIS, Remote Sensing & Geospatial Analysis

- ▶ ArcGIS, QGIS, Google Earth Engine (GEE), ERDAS Imagine
- ▶ AHP, TOPSIS for groundwater potential mapping, suitability assessment and LULC change detection
- ▶ HEC-HMS, HEC-RAS, MODFLOW, GMS for hydrological and groundwater modelling

v. Programming & Machine Learning

- ▶ Python, Google Colab; Random Forest, XGBoost, SVM, SHAP, CA-Markov modelling

vi. Sediment Characterization & Marine Studies

- ▶ Marine sediment sampling, grain size analysis, heavy mineral and organic matter characterization

■ PROFESSIONAL RECOGNITION & AWARDS

- ▶ **Best Researcher Award in Earth and Planetary Sciences — Book of Award, September 2025**
- ▶ Associate Student Member (ID: 145289), International Association of Hydrogeologists (IAH) — since August 2024

■ PEER-REVIEWED JOURNAL PUBLICATIONS (SCI/SCOPUS/WOS INDEXED)

1. Abishek, S.R., Pitchaimani, V.S., Ravindran, A.A., R. Sakthi Priya, S. Kaliraj, **A. Antony Alosanai Promilton** & R. J. Jerin Joe. Integrated assessment of soil and rock penetration across the crystalline basement of the lower Thamirabarani river basin, southern India. *Discover Civil Engineering*, 3, 68 (2026). 10.1007/s44290-026-00451-9
2. Pitchaimani, V.S., Joe, R.J.J., Stuvur, S.S.R., Liberthin, M., **Promilton, A.A.A.** & Abishek, S.R. Landslide susceptibility and vulnerability assessment using GIS-AHP in the highland of Wayanad, Western Ghats, India. *Discov. Hazards*, 2, 5 (2026). 10.1007/s44475-026-00008-5
3. Pitchaimani, V.S., Abishek, S.R., Joe, R.J.J. & **Promilton, A.A.A.** Seasonal variations of inner shelf sediment dynamics and granulometric characteristics along the Nagapattinam coast of the Cauvery basin, Southern India. *Discover Oceans*, 3, 4 (2026). 10.1007/s44289-025-00109-x
4. Abishek, S.R., Pitchaimani, V.S., Kaliraj, S., Priya, R.S., Ravindran, A.A., **Promilton, A.A.A.** & Karuppannan, S. Hydrogeochemical variability and health risk assessment of heavy metals in groundwater of Srivaikundam region, Southern India. *Environmental Geochemistry and Health*, 48, 37 (2026). 10.1007/s10653-025-02914-x

5. Grace, C.A.Z., Soundranayagam, J.P., Sidhu, N., et al. & **Promilton, A.A.A.** Decadal dynamics of urban heat Islands in a coastal industrial region of India: remote sensing insights for sustainable urban planning. *Environment, Development and Sustainability*, (2025). 10.1007/s10668-025-07153-9
6. Ramachandran, A., **Promilton, A.A.A.**, Pitchaimani, V.S., et al. Provenance and Coastal Dynamics inferred through Textural parameters and Heavy Mineral contents of the Shoreline Sediments, Tamil Nadu, Southeast India. *J. Indian Assoc. Sedimentologists*, 42(II), 34–51 (2025). 10.51710/jias.v42il.445
7. Grace, C.A.Z., Soundranayagam, J.P., et al. & **Promilton, A.A.A.** Quantifying spatiotemporal LULC change and urban expansion using geospatial modelling and Shannon's entropy in a coastal city of India. *Discover Civil Engineering*, 2, 189 (2025). 10.1007/s44290-025-00341-6
8. Joe, R.J.J., Pitchaimani, V.S., Kaliraj, S., Abishek, S.R., **Promilton, A.A.A.** & Karuppannan, S. Spatial modelling of groundwater potential zones in the Neyyar Basin using machine learning and morphometric analysis. *Scientific Reports*, 15, 36171 (2025). 10.1038/s41598-025-20000-1
9. **Promilton, A.A.A.**, Ravindran, A.A., Pitchaimani, V.S., Kingston, J.V., Abishek, S.R. & Joe, R.J.J. Characterization of Fractured Basement Aquifers Using Square Array VES, Azimuthal Resistivity Analysis and 3D Lithological Modelling: An Integrated Approach for Sustainable Groundwater Management. *Water Conservation Science and Engineering*, 10, 114 (2025). 10.1007/s41101-025-00444-2
10. Grace, C.A.Z., Soundranayagam, J.P., **Promilton, A.A.A.**, et al. Integrating Remote Sensing and Geospatial-Based Comprehensive MCDA Approach for Sustainable Coastal Solar Site Selection in Southern India. *ISPRS Int. J. Geo-Information*, 14, 377 (2025). 10.3390/ijgi14100377
11. **Promilton, A.A.A.**, Ravindran, A.A., Pitchaimani, V.S., Kingston, J.V. & Karuppannan, S. Comprehensive hydrogeochemical characterization and seasonal water quality index analysis for sustainable groundwater management in Valliyur region, Southern Tamil Nadu, India. *Scientific Reports*, 15, 33251 (2025). 10.1038/s41598-025-18285-3
12. Abishek, S.R., Ravindran, A.A., Pitchaimani, V.S., S. Kaliraj, J. Vinoth Kingston, **A. Antony Alosanai Promilton** & Joe, R.J.J. Hydrogeophysical characterization of groundwater and subsurface geology using VES and GMS modeling in Srivaikundam region, Southern Tamil Nadu. *Modeling Earth Systems and Environment*, 11, 339 (2025). 10.1007/s40808-025-02542-5
13. Grace, C.A.Z., Soundranayagam, J.P., **Promilton, A.A.A.** & Pitchaimani, V.S. Mapping of Cyclone Disaster Vulnerability using Geospatial Techniques and AHP: Implications for Resilient Urban Planning in Coastal City of Southern Tamil Nadu. *Disaster Advances*, 18(7), 50–60 (2025). 10.25303/187da050060
14. Aswin, S.K., Pitchaimani, V.S. & **Promilton, A.A.A.** Flood Susceptibility Assessment for Coastal Villages of Southern Tamil Nadu: An Integrated GIS and AHP Approach. *Disaster Advances*, 18(6), 18–27 (2025). 10.25303/186da018027
15. Priya, R.S., Ravindran, A.A., Abishek, S.R., et al. & **Promilton, A.A.A.** An application of remote sensing and GIS used in groundwater potential zones for sustainable urban development in coastal areas between Chettikulam and Kolachal, southern India. *Applied Geomatics*, (2025). 10.1007/s12518-024-00604-4
16. Grace, C.A.Z., Soundranayagam, J.P., **Promilton, A.A.A.** & Pitchaimani, V.S. Synergizing Hazard Resilience And Renewable Energy Potential For Smart Coastal Development: A GIS-MCDA Approach In Thoothukudi, India. *Nanotechnology Perceptions*, 20(S13), 2608–2628 (2024).
17. Ravindran, A.A., **Promilton, A.A.A.**, Kingston, J.V., Abishek, S.R., Abinaya, R. & Priya, R.S. Multicriteria Analysis of Groundwater Potential in Valliyur Region, Tamil Nadu: Integrating Remote Sensing, GIS and AHP. *Water Conservation Science and Engineering*, 9, 53 (2024). 10.1007/s41101-024-00285-5
18. Ravindran, A.A., Abinaya, R., Abishek, S.R., Kingston, J.V., **Promilton, A.A.A.** & Priya, R.S. Geophysical exploration of tectonic signatures in proterozoic quartzite of the Kovilpatti region, Southern India. *Carbonates and Evaporites*, 39, 59 (2024). 10.1007/s13146-024-00962-5
19. Ravindran, A.A., **Promilton, A.A.A.**, Kingston, J.V., et al. Geophysical and geochemical studies on sinking stream occurrence in Ayankulam village of South Tamilnadu, India. *Proc. Indian National Science Academy*, (2024). 10.1007/s43538-024-00287-w
20. Ravindran, A.A. & **Promilton, A.A.A.** Geophysical and geochemical prospecting for submarine groundwater discharge in a karst leaky aquifer from Valliyur Hills to Uvari Beach. *Modeling Earth Systems and Environment*, 8, 4909–4921 (2022). 10.1007/s40808-022-01412-8
21. Kingston, J.V., Ravindran, A.A., Abishek, S.R., Aswin, S.K. & **Promilton, A.A.A.** Integrated geophysical and geochemical assessment of submarine groundwater discharge in coastal terrace of Tiruchendur, Southern India. *Applied Water Science*, 12, 9 (2022). 10.1007/s13201-021-01553-8
22. Jeyapaul, V.K., Antony Jebamalai, A.R., Selvam, R.A., Krishnaveni, A.S. & **Antony Johnson, A.A.P.** A case study of freshwater discharge in porous calcarenite formation in coastal terrace at Manapad, South India. *Carbonates and Evaporites*, 35, 114 (2020). 10.1007/s13146-020-00641-1

■ BOOK CHAPTERS & CONFERENCE PROCEEDINGS

1. Rajendran, A., Ravindran, A., Promilton, A.A., Ganesan, S. 'Interacting RS and GIS to GWPZ in Urban Development Site Suitability Analysis of Kovilpai Region, Southern Tamil Nadu, India.' (2025)
2. Promilton, A.A., Ravindran, A., Afzal, M., Rajendran, A. 'Groundwater Exploration in Fractured Charnockite Formation for Irrigation at Alangulam, Tamil Nadu, India.' (2024)
3. Ravindran, A., Kingston, J.V., Samy, M., Promilton, A.A., Rajendran, A. 'Groundwater Management Studies Using Geological and 2D ERI Techniques in Ariyanayakupuram Village, Tirunelveli, Tamil Nadu, India.' (2023)
4. Abishek, S.R., et al. 'Groundwater Exploration Study using 2D ERI and Geochemical Mapping in Muruganpuram, Vallanadu, Tamil Nadu.' CESS-2022, pp. 66–70.
5. Kingston, J.V., et al. 'Freshwater discharge in coastal aquifer study using electrical resistivity and geochemical method in Sahupuram, Thoothukudi.' CESS-2022, pp. 50–56.
6. Kanagaraj, R., et al. 'Crystalline limestone investigation study in Podupatti and Pungarvattam, Ettayapuram, Thoothukudi.' CESS-2022, pp. 134–138.
7. Promilton, A.A., et al. 'Crystalline limestone investigation study in Podupatti and Pungarvattam, Ettayapuram, Thoothukudi.' CESS-2022, pp. 134–138.

■ PAPER PRESENTATIONS

International Conferences

- ▶ 'Signature of Sinking Streams on Calcarene Rock Study Using Geophysical and Geochemical Investigation in Ayankulam Area, South Tamil Nadu, India' — ICRAWST-2022, December 2022
- ▶ 'Exploration of Ground Water Using Azimuthal Square Array Method and Magneto Telluric Method in Quartzite Rich Terrain of Pudukkottai, Tamil Nadu' — CESS-2022

■ JOURNAL REVIEWER SERVICE

- ▶ Environmental Monitoring and Assessment
- ▶ Groundwater for Sustainable Development
- ▶ Sustainable Water Resources Management

■ REFERENCES

1. Dr. A. Antony Ravindran

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Declaration: I hereby declare that the information provided above is correct to the best of my knowledge and belief.

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